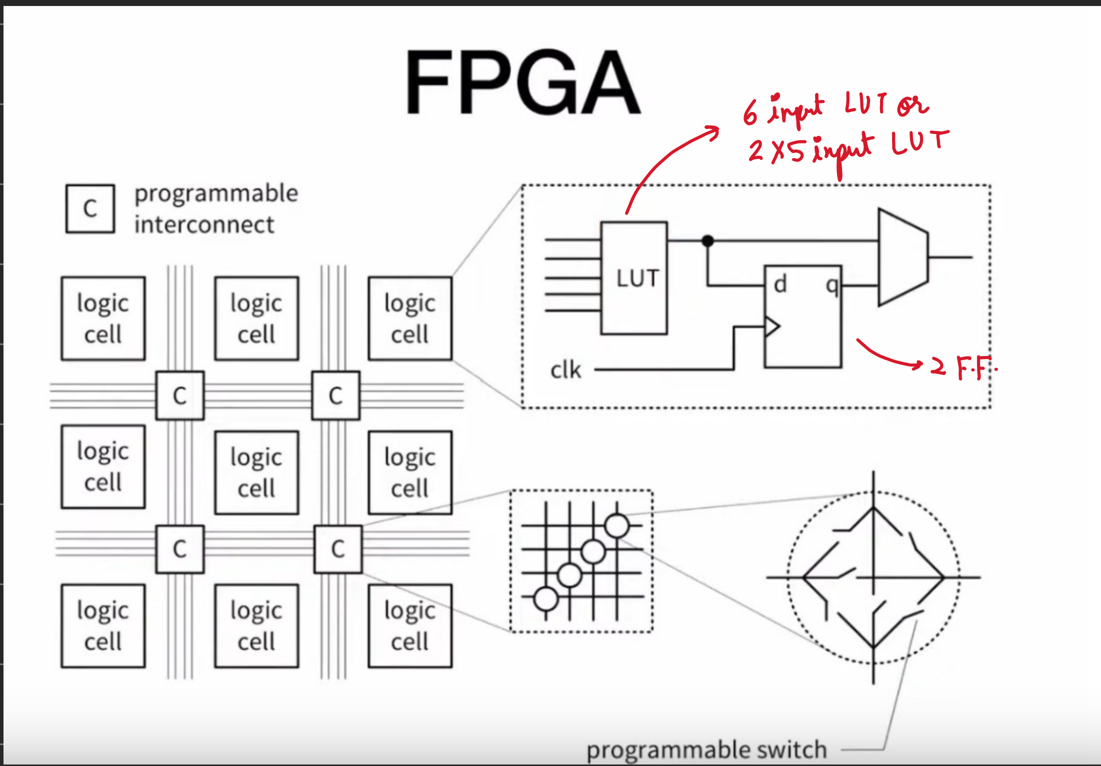
Artix-7 Resources



if want only comb. circuit, no F.F. will be utilized.

macro cells → non configurable, they are pre-implemented memory, clock management circuit

4 Logic Cells (LC) = Slice

2 Slice = Configurable Logic Block (CLB)

Artix-7

Table 1-2: **Artix-7 FPGA CLB Resources**

	Device	Slices[1]	SLICEL	SLICEM	6-input LUTs	Distributed RAM (Kb)	Shift Register (Kb)	Flip-Flops
	7A12T	2,000[2]	1,316	684	8,000	171	86	16,000
	7A15T	2,600[2]	1,800	800	10,400	200	100	20,800
	7A25T	3,650	2,400	1,250	14,600	313	156	29,200
Basys 3	7A35T	5,200[2]	3,600	1,600	20,800	400	200	41,600
Nexys A7	7A50T	8,150	5,750	2,400	32,600	600	300	65,200
	7A75T	11,800[2]	8,232	3,568	47,200	892	446	94,400
Nexys A7	7A100T	15,850	11,100	4,750	63,400	1,188	594	126,800
	7A200T	33,650	22,100	11,550	134,600	2,888	1,444	269,200

Notes:

1. Each 7 series FPGA slice contains four LUTs and eight flip-flops; only SLICEMs can use their LUTs as distributed RAM or SRLs.
2. Number of slices corresponding to the number of LUTs and flip-flops supported in the device.

$15850 \times 4 = 63{,}400 \times 2 = 126{,}800$

Artix-7 Memory Types

	Device	Slices[1]	SLICEL	SLICEM	6-input LUTs	Distributed RAM (Kb)	Shift Register (Kb)	Flip-Flops
Nexys A7	7A100T	15,850	11,100	4,750	63,400	1,188	594	126,800

Distributed RAM	Block RAM (BRAM)
Constructed from LUT	Prefabricated as Macro Cells
Competes with normal logic for resources	Separated from logic cells (LC)
Asynchronous	Wrapped by synchronous, configurable interface
Xilinx Only	Found in most modern FPGA devices

BRAM Size

- Each BRAM Macro Cell has
  - 32K (2^{15}) data bits
  - 4K (2^{12}) parity bits
  - ↳ Total of 32K + 4K = 36K bits

- BRAM can be arranged in different widths/depths.
  - 32K by 1 ($2^{15} \times 1$)

    ⋮
  - 512 by 64 ($2^9 \times 2^6$)

- BRAM can be partitioned into 2 independent 18 kb BRAMs.

- Artix-7 XCA100T has 135 BRAM macros (totaling 135 × 32K = 4320 kb usable data bits.)